Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION DECEMBER – 2025

Name of the course: B. Tech
Name of Paper: Material science
Time: 3 Hours

Semester: 3rd
Paper code: TME302
Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Why is the study of Materials Science and Engineering important for a Mechanical Engineer? Explain with examples.	CO1, CO2
(b)	Explain the basic types of interatomic bonds with suitable examples.	
(c)	Explain how X-ray diffraction is used to determine crystal structure. Derive Bragg's Law.	
Q2	(20 marks)	
(a)	Describe the crystal structures of NaCl, CsCl, Diamond cubic, and Zinc blende.	CO2, CO3
(b)	Classify materials based on structure and properties.	
(c)	What are imperfections in solids? Classify and explain them.	
Q3	(20 marks)	
(a)	Determine the lattice parameter of a cubic crystal if the first-order Bragg reflection from the (200) plane occurs at 44.5° using Cu Kα radiation (λ = 1.54 Å).	CO3, CO4
(b)	Describe the different types of point defects found in crystals.	
(c)	Explain Fick's First and Second Laws of diffusion and their significance.	
Q4	(20 marks)	
(a)	Discuss the features of the Fe–C phase diagram and mark all important transformations.	C04, CO5
(b)	Explain the Al–Cu and Al–Si systems and their industrial significance.	
(c)	For a steel containing 0.4 % C, determine the weight percentage of ferrite and cementite just below the eutectoid temperature using the lever rule.	
Q5	(20 marks)	
(a)	Discuss various heat treatment processes such as annealing, normalizing, hardening, and tempering and their effect on microstructure and properties.	C05 CO6
(b)	Explain the stress–strain behaviour of metals, ceramics, and polymers	
(c)	Explain the free electron theory and derive an expression for Fermi energy.	